

Causal Delay Copilot

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The Problem

Large capital construction projects typically take 20% longer to finish than scheduled and up to 80% over budget. **Across megaprojects worldwide, 98% run over budget by more than 30%, and 77% run at least 40% behind schedule**, according to [McKinsey research](#). These numbers show up on almost every large project, and the reason they keep repeating is simple. Teams usually find out that a shipment or a task is at risk. They rarely find out why.

AI tools built for construction supply chains, including current predictive platforms, are designed to spot this risk early. They look at patterns in historical data and flag when something looks likely to go wrong, which is genuinely useful. But when a project manager sees that flag, they are still left guessing at the actual cause and what to do about it, usually under time pressure and with incomplete information.

Our Solution: Causal Delay Copilot

We are building a system that adds the layer current tools are missing: causal explanation. Instead of only flagging that a shipment is at risk, it identifies which specific factor is actually driving the delay, estimates how much that factor contributes in days, and hands the project manager one concrete action to take.

The core question our demo answers is direct. Does relying on a supplier who is also serving other active projects causally increase delivery delay, once material type, order size, and season are accounted for. The system runs this analysis on order and delivery data, surfaces the true driver behind a flagged risk, and passes it to a lightweight agent that drafts one action, such as an expedite request or an alternate vendor suggestion, tied directly to that driver.

What Makes This Different

Most AI tools in this space, including existing predictive supply chain platforms, are correlational. They are good at telling a team that a delay is coming, based on patterns learned from past data. What they are not built to do is separate a true cause from something that simply moves together with delay in the data.

Causal Delay Copilot is built specifically to answer that second question. It is not trying to out-predict existing tools. It sits on top of prediction and turns a risk flag into a reason and a recommended action, which is the part of the workflow a project manager actually needs in order to act quickly.

The Technique

The core of the system is causal machine learning, specifically Double Machine Learning and Generalized Random Forests, implemented using EconML. Unlike a standard predictive model, this class of technique is designed to estimate the true effect of one specific factor, such as a shared supplier, on an outcome like delivery delay, while properly separating out confounding factors like order size or season that a correlational model would otherwise mix in.

This is not a new or unproven approach. Recent research by [Wyrembek, Baryannis, and Brintrup \(2024\)](#) has shown that causal machine learning can isolate specific, actionable drivers of delay in supply chain settings, including cases where sharing a supplier across multiple buyers was linked to a measurable increase in delivery delay once other factors were controlled for. We apply the same class of technique to construction procurement and scheduling data, adapted to the delay drivers construction teams deal with day to day, such as shared vendors, material type, and seasonal exposure.

Closing Note

Causal Delay Copilot is proposed as a complementary layer that sits on top of existing predictive supply chain platforms, not a replacement for them. It is meant to work alongside the prediction and automation these platforms already provide, adding the causal explanation and the recommended action that turns a risk flag into a decision a project manager can act on right away.